

Nila Mönig, PhD

moenigin@gmail.com | Freiburg (DE) | German | [linkedin.com/in/nilamoenig](https://www.linkedin.com/in/nilamoenig) | github.com/moenigin

PROFILE

PhD neuroscientist turning large, complex image datasets into reliable, reproducible results. Develops deep-learning pipelines (PyTorch, HPC) together with the tools, quality control and documentation around them, and works across biology, engineering and external partners. Hands-on from hardware to code: builds solutions where no standard tool exists.

CORE COMPETENCES

Computational Microscopy: Electron microscopy | Segmentation | Object classification | Feature extraction

Machine Learning & AI: PyTorch | MONAI | Gunpowder | scikit-learn | U-Net / ResNet architectures

Workflow & Reproducibility: Automation | Versioned pipelines | Documentation | Quality control

Scalable data processing: SLURM/HPC | Array-based deployment | Parallel inference | Process automation

Data Analytics: Statistical analysis | Multi-dimensional data analysis | Clustering | Dimensionality reduction

Collaboration & Training: Mentoring | Documentation | Cross-disciplinary communication | Presentation

PROFESSIONAL EXPERIENCE

Independent Work, JUL 2026 – present

- Finalising the synapse-detection pipeline (code cleanup, documentation, validation) for release
- Revising a manuscript for journal publication

POSTDOCTORAL SCIENTIST

Friedrich Miescher Institute for Biomedical Research, Basel – Oct 2022 – Jun 2026

Led development of scalable data pipelines for connectomics analysis in zebrafish neuroscience

- **Designed and implemented** a high-throughput synapse-detection pipeline tailored to non-standard tissue properties.
- **Developed, trained, and deployed** deep-learning segmentation and classification models (U-Net, ResNet variants) on HPC infrastructure using array-based parallel deployment.
- **Created and maintained** custom proofreading and training-data tools in napari and Neuroglancer; **authored** documentation and integrated user feedback.
- **Coordinated** external proofreading partnerships, **established** benchmarking protocols, and **performed** quality control across platforms and data formats.
- **Designed** the experimental and computational strategy for the connectomics component; **advised** on data quality and segmentation readiness.
- **Mentored** colleagues in connectomics terminology and data-generation standards. **Supervised and mentored** an intern in data annotation and tool use.
- **Evaluated** image quality and **provided feedback** to acquisition teams to support segmentation readiness for datasets of different origins.
- **Contributed** to a collaborative, inclusive research environment and knowledge-exchange culture in a diverse international team.

PHD CANDIDATE

Friedrich Miescher Institute for Biomedical Research - May 2012 - Sep 2022

Thesis: Neuronal Organisation of the Olfactory Bulb in Adult Zebrafish | magna cum laude (5.5/6.0)

Developed integrated connectomics and behavioural-analysis workflows to investigate neural-circuit function in the zebrafish olfactory system.

- **Partnered** with Google Research on neuron segmentation for TB-scale EM datasets, translating biological requirements into technical specifications for machine learning model development
- **Generated** ground truth training datasets, developing deep understanding of computer vision model requirements and data quality standards
- **Developed** a neuroglancer-based proofreading interface that enabled efficient inspection and curation of automated segmentations, substantially accelerating reconstruction workflows (**15-50× speed improvement**).
- **Proofread** >450 neurons and **established** a comprehensive morphological classification scheme.
- **Trained** a student assistant on neuron proofreading
- **Co-developed and automated** behavioural assays, reducing manual work by 87% while doubling throughput; **implemented** a multi-dimensional analysis pipelines (Python/MATLAB).
- **Designed and built** custom behavioural apparatus (prototyping, fabrication, fluidics, control electronics)
- **Contributed** behavioural training and analysis to cross-disciplinary projects linking neural function and behaviour.

RESEARCH ASSISTANT, University of Freiburg - Jan 2011 - Mar 2012

- **Prepared and maintained** primary neuronal cell cultures ensuring viability and experimental quality.
- **Implemented** closed-loop stimulation circuits (Tucker Davis Technologies).
- **Performed** electrophysiology, pharmacology, immunohistochemistry and neuron tracing.

STUDENT ASSISTANT, University of Freiburg - Jun 2007 - Sep 2009

- **Teaching assistant** for introductory MATLAB course.
- **Contributed** to data-management system development and neuroanatomical analyses.

EDUCATION

- PhD Neuroscience, University of Basel (2022) - magna cum laude
- Diplom Biology, University of Freiburg (2009) - 1.0 (best possible)

SELECTED PUBLICATIONS & PRESENTATIONS

- Mönig N.R*, Januszewski M. * et al., 2026, **bioRxiv** 2026.07.13.738197 (preprint)
- Frank T., Mönig N.R., Satou C. et al., **Nat Neurosci** 22: 1844-1856 (2019)
- Namekawa I.*, Mönig N.R*, **Exp Brain Res** 236: 2959-2969 (2018)

Poster and oral presentations at international meetings (CONNECTOMICS, FENS, ECRO, ZEBRAFISH BRAIN CONFERENCE) and local neuroscience meetings. (*equal contribution)

SELECTED COURSES & TRAINING

- Entrepreneurship Training, Innosuisse Module 2 (Innovation Office, University of Basel, 2026)
- Advanced Python and Machine Learning (University of Basel, 2024)
- Dealing with Research Data Management (University of Basel, 2024)
- Deep Learning for image analysis (EMBL, Heidelberg, 2023)
- antelope@novartis (University of Basel, 2017)

LANGUAGES

German (native), English (C2), French (B1)

HOBBIES

- running
- furniture upcycling
- knitting